

Jio Subscriber Retention & AI Chatbot Project

Day 2 Completion Report | Predictive Modelling | Project-5 | Submitted 21 Sep 2026

Scope

Day 2 milestone per the homework brief: plan and build predictive models for (1) 30-day customer churn, (2) Customer Lifetime Value (CLV), and (3) offer-response / uplift - who will actually change their mind if given a discount, as opposed to who was never going to leave. Logistic Regression was used as the baseline throughout, with Gradient Boosted Trees (XGBoost, CatBoost) as champion candidates, per the homework's specification. All three models were trained on the Day 1 feature set (*data/processed/features.csv*, 64,738 subscribers, 70 features after encoding), with the three confirmed leakage columns (*mnq_enquiry_flag*, *churn_reason*, *churn_date*) excluded throughout.

1. 30-Day Churn Model

Target: *churn_flag_30d* (1.73% positive rate). Evaluated on 5-fold cross-validated PR-AUC rather than accuracy, since a model predicting "nobody churns" would score 98.3% accuracy while being useless.

Model	CV PR-AUC	Recall@top-decile*
Logistic Regression (baseline)	0.207	0.746
CatBoost	0.201	-
XGBoost	0.178	0.670

*Recall@top-decile measured on a single stratified 80/20 split; CV PR-AUC is the primary, more stable comparison metric.

Decision: Logistic Regression selected as the final churn model. It won on every metric tried, against both XGBoost and CatBoost, even after tuning *scale_pos_weight* and validating with 5-fold CV rather than a single split. This is a legitimate, reportable finding rather than a modelling failure: it indicates the churn signal in this dataset is close to linearly separable in the engineered feature space - consistent with the smooth, monotonic relationships found in the Day 1 diagnostics (e.g. the steady ARPU-band and tenure-bucket churn curves) - so added tree-ensemble complexity did not translate into better ranking of at-risk subscribers.

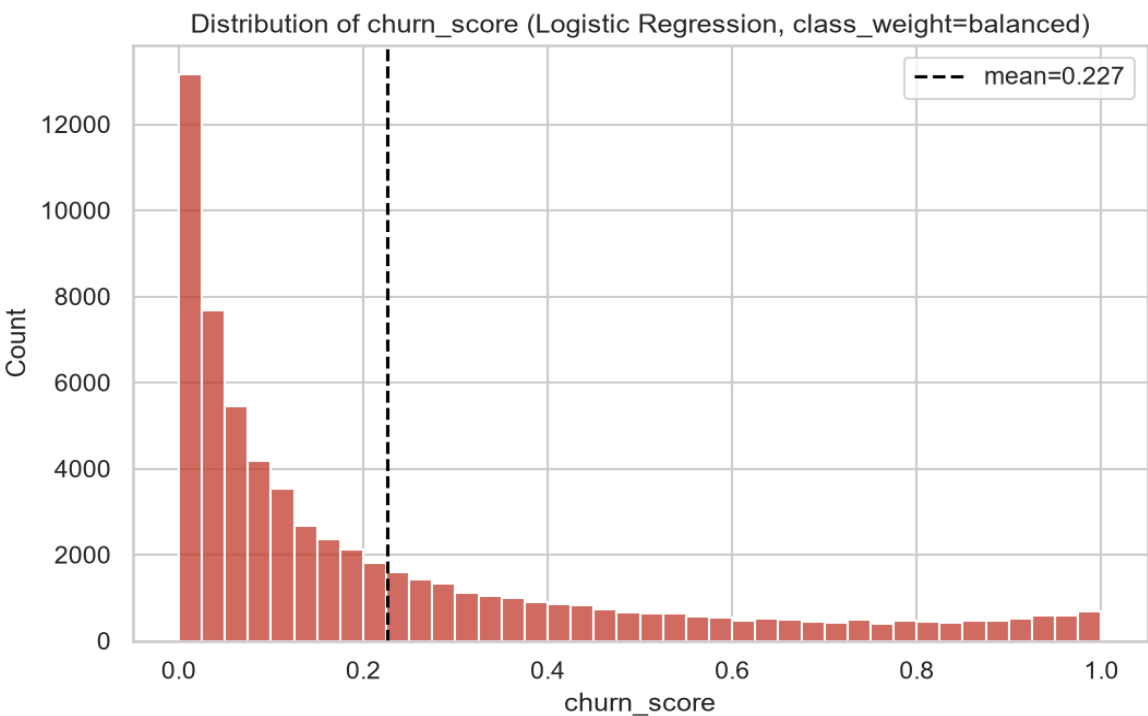


Chart 9 - Distribution of the final churn_score across all 64,738 subscribers

Calibration caveat: because the model was trained with `class_weight="balanced"` to counter the 1.73% event rate, the resulting scores are not calibrated probabilities - mean churn_score is 0.227, far above the true 30-day base rate. The score is reliable for **ranking** subscribers (top-N, deciles, segment comparisons) but should not be read as "this subscriber has a 61% chance of churning." This is flagged explicitly so the Day 3-4 chatbot and any business consumer of this score doesn't misinterpret it.

2. Customer Lifetime Value (CLV)

Target constructed as $\text{arpu_6m_avg_inr} * 12$ - a simple 12-month value projection, stated here as an explicit assumption. All ARPU-derived columns ($\text{arpu_last_month_inr}$, arpu_3m_avg_inr , arpu_6m_avg_inr) were excluded from the feature set so the model could not trivially reconstruct the target from its own input.

Model	MAE (Rs)	R2
Linear Regression (baseline)	614.7	0.584
Random Forest (champion)	610.1	0.580

Decision: essentially a statistical tie. Random Forest edges out on MAE, Linear Regression edges out on R2 - neither difference is large enough to declare a clear winner. An R2 of ~0.58 using only behavioural features (no ARPU leakage) is a solid, realistic result: roughly 58% of the variance in projected annual value is explained by recharge, usage, tenure and service-quality behaviour alone.

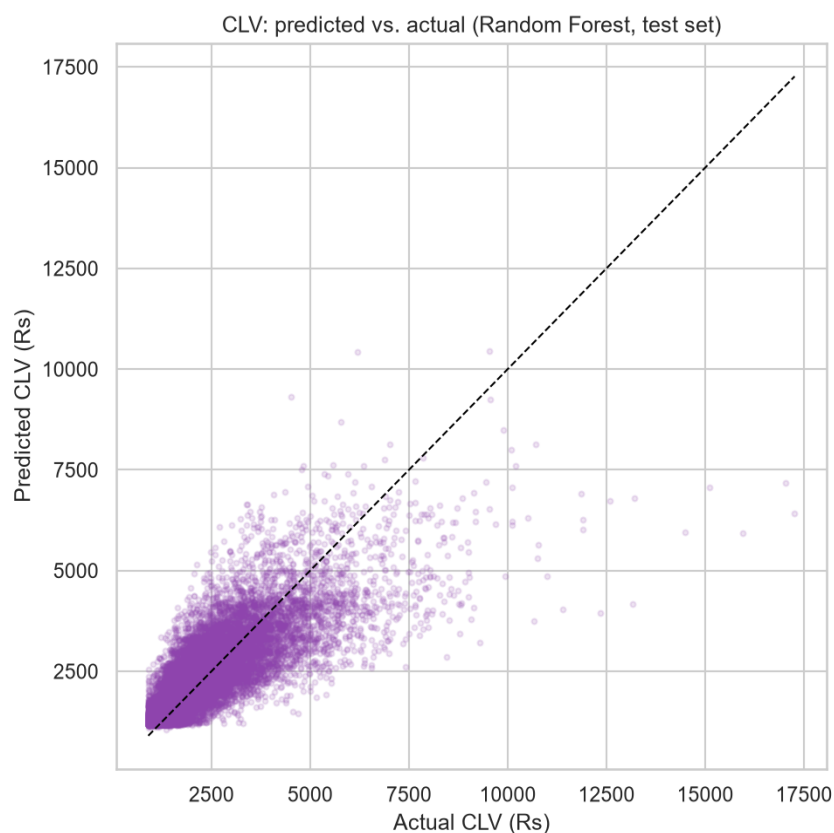
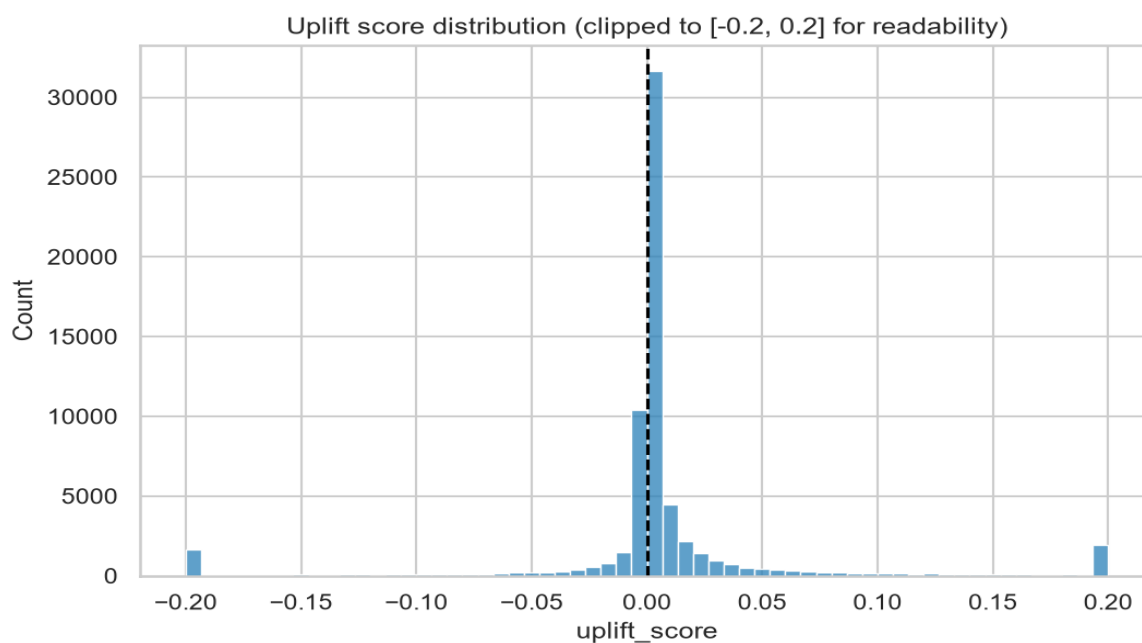
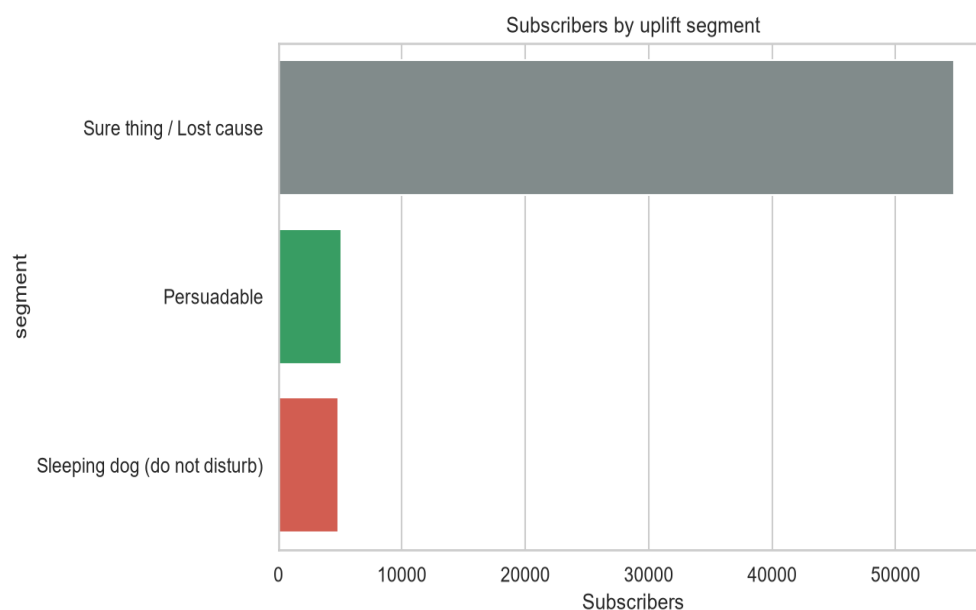


Chart 12 - CLV predicted vs. actual (Random Forest, held-out test set). Points hugging the diagonal would indicate a perfect fit; the visible spread reflects the realistic ~42% of variance not explained by behavioural features alone.

3. Offer-Response / Uplift Model

Two-model (T-learner) approach: separate XGBoost classifiers trained on the treated (offer_exposed_90d = 1, n = 13,991) and control (n = 50,747) populations, each predicting retention (1 - churn_flag_90d). Uplift score = $P(\text{retain} \mid \text{treated}) - P(\text{retain} \mid \text{control})$ for every subscriber.

Segment	Subscribers	% of base	Interpretation
Sure thing / Lost cause	54,751	84.6%	Outcome unaffected by offer
Persuadable	5,118	7.9%	Offer changes their outcome - target these
Sleeping dog	4,869	7.5%	Offer increases churn risk - exclude these



Decision: this is the model that directly operationalises the engagement brief's core argument (section 6.3) - that propensity-only targeting wastes spend on subscribers who were never going to leave. Only ~8% of the base is genuinely persuadable by an offer; retention spend should concentrate there. The 7.5% "sleeping dog" segment should be actively **excluded** from campaigns, since an offer measurably increases their churn risk

rather than reducing it - a real and counter-intuitive effect that pure propensity scoring would miss entirely.

Caveat: the score distribution has a very tight interquartile range (-0.00004 to 0.0061 - most subscribers are barely moved either way) but fat tails (min -0.97, max +0.91). This is a known limitation of the two-model approach when the treated and control populations don't fully overlap in feature space, producing extrapolated, less reliable scores at the extremes. Noted rather than corrected, given the scope of this exercise.

4. Combined Scored Table

All three models' outputs were merged into `data/processed/scored_subscribers.csv` (64,738 rows: `subscriber_id`, `churn_score`, `clv_predicted`, `uplift_score`, `segment`). This is the exact table to be imported into SQL Server on Day 3, so the GenAI chatbot can answer questions such as "why is this subscriber high risk, and what should we offer" by joining subscriber identity to all three model outputs in one place.

Key Takeaways

- Simplicity won on two of three models: Logistic Regression beat both gradient-boosted candidates on churn, and tied Random Forest on CLV - a direct, honestly-reported consequence of this dataset's largely linear/monotonic relationships (also visible in Day 1's diagnostics).
- The churn score is a ranking tool, not a calibrated probability - this must be communicated clearly to any business user or chatbot consumer of the score.
- Uplift modelling is the highest-value addition of the three: it identifies both who to spend retention budget on (7.9% persuadable) and who to actively leave alone (7.5% sleeping dogs) - neither of which a simple churn-risk score alone would reveal.
- CLV at $R^2 \sim 0.58$ is usable for ranking subscribers by value-at-risk when combined with the churn score, even though it doesn't perfectly predict exact rupee value.

Next Steps (Day 3 Preview)

Import `subscribers`, `service_requests`, `network_sites`, `circle_monthly_kpi`, `circle_targets`, `offer_catalogue`, and this `scored_subscribers` table into SQL Server via SSMS; build the schema catalog, semantic layer, and few-shot query bank the Text-to-SQL agent will need.

Sources: `data/processed/scored_subscribers.csv`, `uplift_scores.csv`, `clv_predictions.csv`, `outputs/charts/09-12`, generated via `scripts/day2_*.py`.